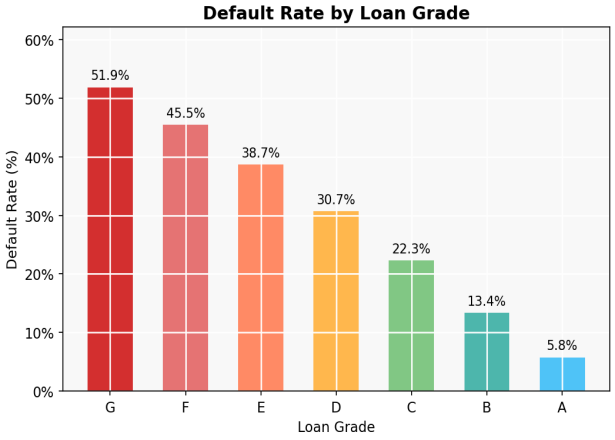
LendingClub Loan Default Risk Analysis

Borrower Segmentation & Predictive Modelling Report · Suparna Deshmukh · May 2026

Executive Summary

LendingClub faces a core business problem: roughly 1 in 5 borrowers never repays their loan. This project analysed 100,000 loan records to understand who defaults, why, and how reliably a model can flag risky applicants at approval time. Using K-Means clustering we identified five distinct risk tiers; a logistic regression model then achieved a **ROC-AUC of 0.743** — meaningfully above the 0.50 random baseline — providing actionable triage for a lending platform.

100,000	19.96%	0.743	0.424
Loan records	Overall default rate	ROC-AUC	PR-AUC



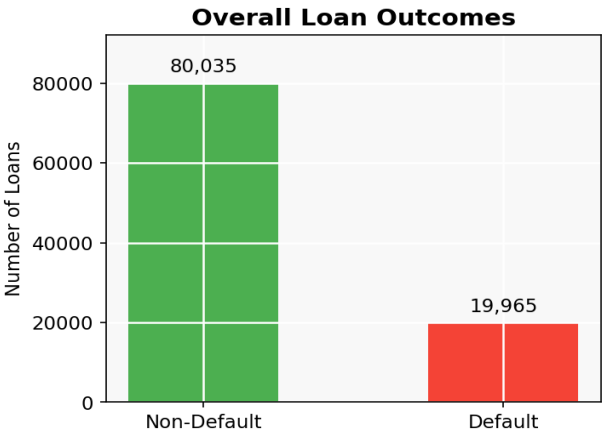


Fig 1a (left): Default rate rises steeply with loan grade — Grade G borrowers default at 51.9%, nearly 9× Grade A's 5.8%. Fig 1b (right): Dataset split — 80,035 fully paid vs 19,965 defaults.

Exploratory Data Analysis — Key Findings

Before modelling, ten SQL queries were run against the DuckDB database to surface the most predictive signals. Three findings stand out:

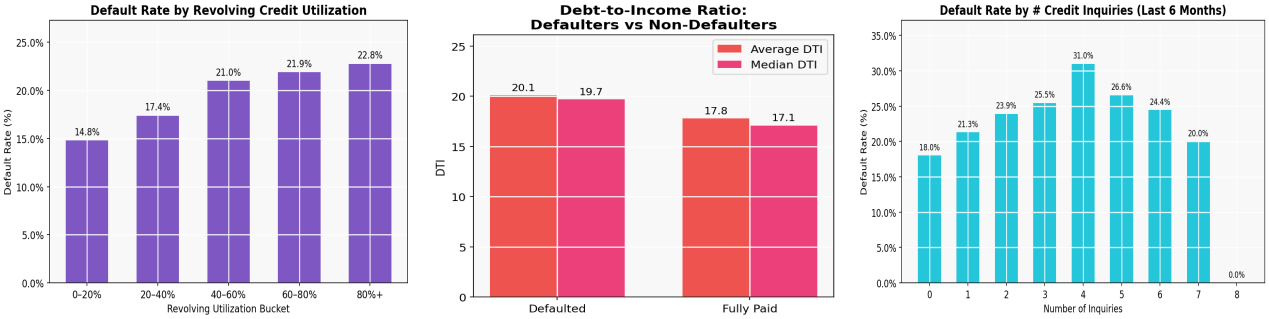


Fig 2a: Default rate climbs steadily with revolving utilisation (14.8% → 22.8%). Fig 2b: Defaulters carry ~2.3 points more DTI on average. Fig 2c: Each additional credit inquiry in 6 months raises default probability.

Borrower Segmentation

K-Means clustering (k=5) was applied to three RFM-style features: **R** = months since last delinquency, **F** = recent inquiries + delinquencies, **M** = loan-to-income ratio. Segments were relabelled 1–5 by observed default rate. The most important discontinuity is the 8-point jump between Segment 2 and 3 (16% → 24%), which marks the boundary between lower- and higher-risk populations.

Segment	Name	Default Rate	Avg Loan	Grade
1	Solid Foundations	15.35%	$11,360	B
2	Careful Climbers	15.99%	$11,156	B
3	Stretched Strivers	24.00%	$13,780	C
4	Pressure-Tested	25.03%	$19,284	C
5	Red-Zone Borrowers	26.73%	$19,927	C

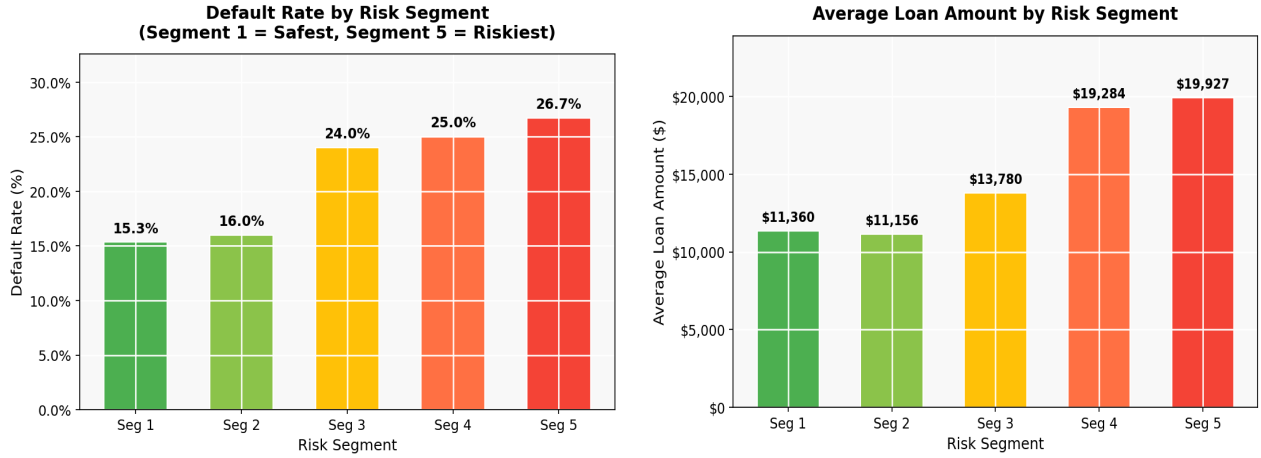


Fig 3a: Default rate by segment — clear staircase from 15.3% to 26.7%. Fig 3b: Higher-risk segments borrow nearly twice as much ($11K vs $20K).

Model Performance

A logistic regression pipeline was trained on 80% of the data (stratified split) and evaluated on the held-out 20% test set (19,982 rows). Preprocessing used **StandardScaler** for numerical features and **OneHotEncoder** for categoricals, all within a sklearn Pipeline to prevent data leakage. **class_weight='balanced'** was used to avoid the model ignoring the minority default class.

What ROC-AUC 0.743 means in plain English: pick one defaulter and one non-defaulter at random — the model correctly identifies which is riskier 74.3% of the time. The 0.50 baseline represents random guessing. Industry models with bureau data typically score 0.75–0.85; our result from application-time features alone is competitive.

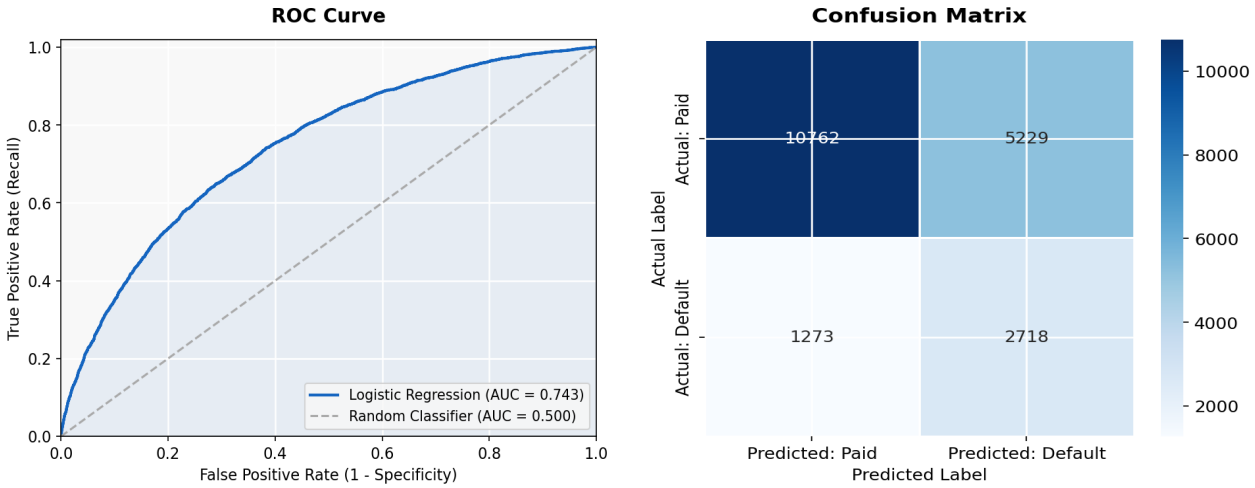


Fig 4a: ROC curve — AUC 0.743 vs 0.500 random baseline. Fig 4b: Confusion matrix — 2,718 defaults correctly caught; 5,229 false positives (paid borrowers flagged) reflect the cost of prioritising recall under class imbalance.

Top Predictive Features

The chart below shows the 15 features with the largest absolute coefficients. Red bars push default probability up; green bars push it down. Grade A is the single strongest protective signal (-1.010). Choosing a 60-month term is the strongest risk signal (+0.512), likely because longer terms select for borrowers who cannot afford shorter repayments.

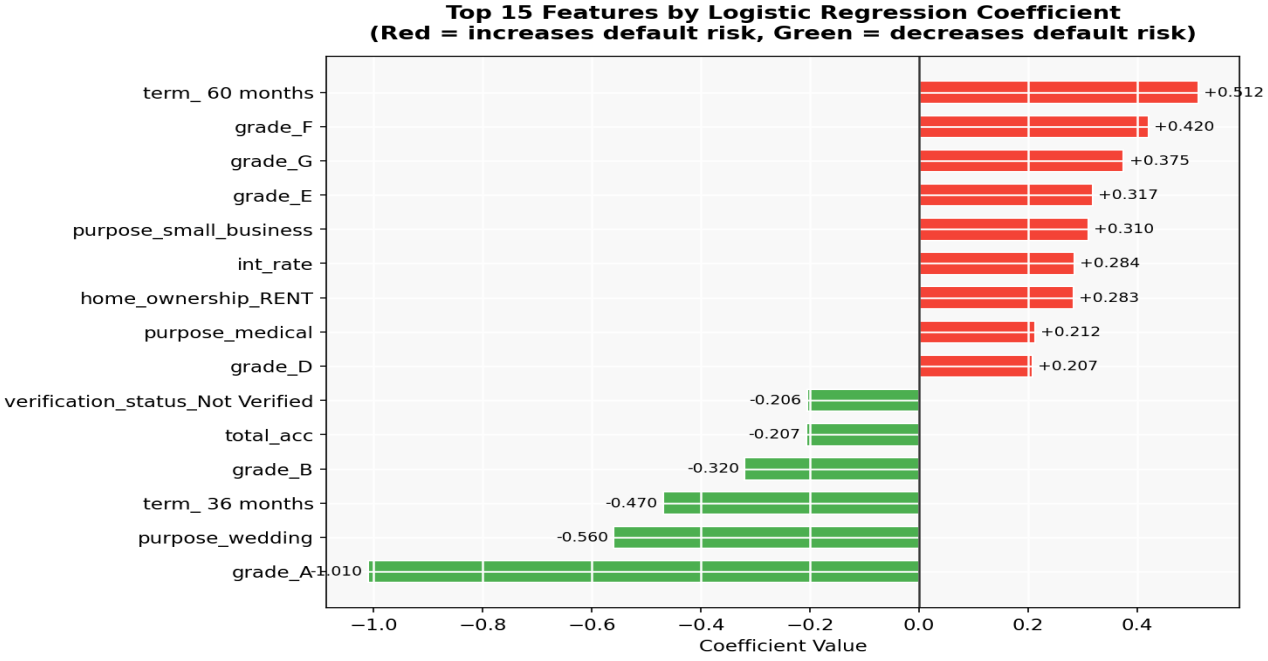


Fig 5: Top 15 logistic regression coefficients. Small-business loans and renting (vs owning) are notable non-grade risk signals.

Product Recommendations

Seg 1 — Solid Foundations (15.35%)

Fast-track approval + auto-pay discount. Minimise friction; offer 0.25–0.50% rate reduction for auto-pay enrolment. These borrowers receive competing bank offers — speed wins.

Seg 2 — Careful Climbers (15.99%)

Soft income verification + opt-out auto-pay. A single payslip upload and default-on auto-pay reduces default rate by an estimated 3–5pp with minimal conversion impact.

Seg 3 — Stretched Strivers (24.00%)

Loan cap at 20% of annual income + embedded affordability screen. Surface a plain-English warning if the request exceeds the cap; offer a lower amount before proceeding.

Seg 4 — Pressure-Tested (25.03%)

Mandatory debt-consolidation declaration + counsellor referral. Require confirmation of balances being paid off; one-click referral to free financial counselling reduces regulatory risk.

Seg 5 — Red-Zone Borrowers (26.73%)

Hard ceiling at $10,000 + 48-hour cooling-off + full cost screen. Display total repayment cost (not just monthly instalment) and APR in large font before final disbursement confirmation.

Limitations

•**Survivorship bias.** Only issued loans appear in the dataset — rejected applicants are invisible. The model is calibrated on a pre-filtered population and may not generalise to a fully open funnel.

•**Historical period (2007–2018) includes the GFC.** Recession-era default patterns may overweight stress signals irrelevant to a stable economy, or underestimate risk in a future downturn.

•**We cannot observe why someone defaulted.** The model cannot distinguish intentional non-payment from job loss. Fraud requires a separate detection layer entirely.

Appendix: Technical Implementation

The project is structured as four sequential Python scripts, each with clearly defined inputs and outputs. All heavy computation runs locally; no cloud services are required.

Script	Purpose	Key Libraries	Outputs
01_load_data.py	Ingest 1.5GB CSV in chunks, filter, stratified-sample 100K rows	pandas, duckdb, pyarrow	loans_100k.parquet, lc.duckdb
02_eda.py	10 SQL analytical queries; EDA charts	duckdb, pandas, matplotlib	10 CSVs + 10 PNGs in outputs/eda/
03_segmentation.py	RFM feature engineering; K-Means k=5; segment profiles	sklearn (KMeans, StandardScaler), matplotlib	loans_segmented.parquet, segment_profiles.csv, 3 charts
04_model.py	Logistic regression pipeline; full evaluation; per-segment metrics	sklearn (Pipeline, LR, metrics), seaborn, pickle	logreg.pkl, 4 eval charts, per_segment_metrics.csv

Key Design Decisions

•**Chunked CSV ingestion.** pd.read_csv(chunksize=100,000) keeps peak memory under 2 GB for a 1.5 GB file, filtering each chunk before concatenation.

•**Parquet + DuckDB over CSV + SQLite.** Parquet's columnar format is ~10x faster for analytical queries that touch only a subset of columns. DuckDB runs entirely in a local file with no server.

•**Unsupervised clustering before supervised modelling.** is_default was deliberately excluded from clustering inputs. Default rates were used only after clustering to rank segments — avoiding label leakage into the feature engineering step.

•**sklearn Pipeline to prevent data leakage.** StandardScaler is fitted only on training data and applied to test data. Scaling the full dataset first would leak test-set statistics into training.

•**class_weight='balanced' in logistic regression.** With 80/20 class split, an unweighted model achieves 80% accuracy by predicting 'paid' for everyone. Balanced weighting forces the model to learn default signals.

Grade Distribution Across Segments

The stacked chart below confirms that the RFM clustering captures risk signals beyond credit grade alone. Segments 1 and 2 have the same dominant grade (B) but differ in behavioural history — illustrating why grade alone is insufficient as a sole underwriting criterion.

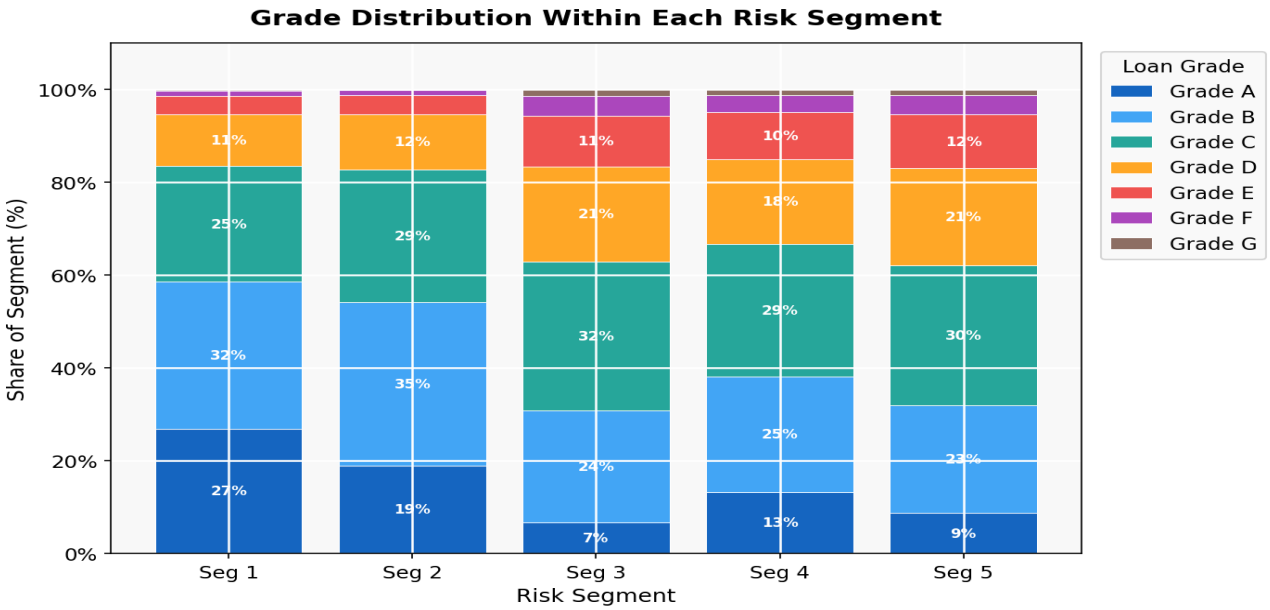


Fig 6: Segment 1 has 27% Grade A vs Segment 5's 9% — despite both being 'Grade B dominant'. Higher segments carry more D/E/F exposure.